1. Design a c++ application.

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit

(globals)

Project Classe Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <string>
#include <sstream>

using namespace std;

// ================= STUDENT CLASS =================
class Student {
private:
    int rollNo;
    string name;
    string course;
    float marks;

public:
    Student(int r, string n, string c, float m) {
        rollNo = r;
        name = n;
        course = c;
        marks = m;
    }

    int getRoll() const { return rollNo; }
    string getName() const { return name; }
    string getCourse() const { return course; }
    float getMarks() const { return marks; }

    void setMarks(float m) { marks = m; }
```

Compiler Resources Compile Log Debug Find Results

Line: 1 Col: 1 Sel: 0 Lines: 156 Lenc Insert Done parsing in 0 seconds

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 6

(globals)

Project Classe Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp

    string getGrade() const {
        if (marks >= 75) return "A";
        else if (marks >= 60) return "B";
        else if (marks >= 50) return "C";
        else return "Fail";
    }

    void display() const {
        cout << rollNo << "\t" << name << "\t"
             << course << "\t" << marks
             << "\t" << getGrade() << endl;
    }

    string serialize() const {
        stringstream ss;
        ss << rollNo;
        return ss.str() + "|" + name + "|" + course + "|" +
               (stringstream() << marks, stringstream().str());
    }
};

// ================= STUDENT MANAGER =================
class StudentManager {
private:
    vector<Student> students;

public:
    void addStudent() {
        int roll;
```

Compiler (24) Resources Compile Log Debug Find Results

Line: 8 Col: 1 Sel: 0 Lines: 156 Lenc Insert Done parsing in 0 seconds

C:\JHIL\jhil application2.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

Project Class Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp
        string name, course;
        float marks;

        cout << "Enter Roll No: ";
        cin >> roll;
        cin.ignore();

        cout << "Enter Name: ";
        getline(cin, name);

        cout << "Enter Course: ";
        getline(cin, course);

        cout << "Enter Marks: ";
        cin >> marks;

        students.push_back(Student(roll, name, course, marks));
        cout << "Student added successfully!\n";
    }

    void displayStudents() {
        if (students.empty()) {
            cout << "No students found.\n";
            return;
        }

        cout << "Roll\tName\tCourse\tMarks\tGrade\n";
        cout << "-----------------------------------------------\n";

        for (size_t i = 0; i < students.size(); i++) {
```

Compiler (24) Resources Compile Log Debug Find Results

Line: 8 Col: 1 Sel: 0 Lines: 156 Leng Insert Done parsing in 0 seconds

C:\JHIL\jhil application2.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

Project Class Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp
            students[i].display();
        }
    }

    void searchStudent() {
        int roll;
        cout << "Enter Roll No to search: ";
        cin >> roll;

        for (size_t i = 0; i < students.size(); i++) {
            if (students[i].getRoll() == roll) {
                students[i].display();
                return;
            }
        }
        cout << "Student not found!\n";
    }

    void updateMarks() {
        int roll;
        float newMarks;

        cout << "Enter Roll No: ";
        cin >> roll;

        for (size_t i = 0; i < students.size(); i++) {
            if (students[i].getRoll() == roll) {
                cout << "Enter new marks: ";
                cin >> newMarks;
                students[i].setMarks(newMarks);
```

Compiler (24) Resources Compile Log Debug Find Results

Line: 8 Col: 1 Sel: 0 Lines: 156 Leng Insert Done parsing in 0 seconds

C:\JHIL\jhil application2.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

Project Class Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp
                cout << "Marks updated!\n";
                return;
            }
        }
        cout << "Student not found!\n";
    }
};

// ================ MAIN ================
int main() {
    StudentManager sm;
    int choice;

    do {
        cout << "\n===== STUDENT RECORD MANAGEMENT =====\n";
        cout << "1. Add Student\n";
        cout << "2. Display All Students\n";
        cout << "3. Search Student\n";
        cout << "4. Update Marks\n";
        cout << "0. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1: sm.addStudent(); break;
            case 2: sm.displayStudents(); break;
            case 3: sm.searchStudent(); break;
            case 4: sm.updateMarks(); break;
            case 0: cout << "Exiting...\n"; break;
            default: cout << "Invalid choice!\n":
```

Compiler (24) Resources Compile Log Debug Find Results

Line: 8 Col: 1 Sel: 0 Lines: 156 Leng Insert Done parsing in 0 seconds

C:\JHIL\jhil application2.cpp - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

Project Class Untitled1.cpp Untitled2.cpp jhil application2.cpp

```cpp

// ================ MAIN ================
int main() {
    StudentManager sm;
    int choice;

    do {
        cout << "\n===== STUDENT RECORD MANAGEMENT =====\n";
        cout << "1. Add Student\n";
        cout << "2. Display All Students\n";
        cout << "3. Search Student\n";
        cout << "4. Update Marks\n";
        cout << "0. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1: sm.addStudent(); break;
            case 2: sm.displayStudents(); break;
            case 3: sm.searchStudent(); break;
            case 4: sm.updateMarks(); break;
            case 0: cout << "Exiting...\n"; break;
            default: cout << "Invalid choice!\n";
        }

    } while (choice != 0);

    return 0;
}
```

Compiler (24) Resources Compile Log Debug Find Results

Line: 8 Col: 1 Sel: 0 Lines: 156 Leng Insert Done parsing in 0 seconds

C:\JHIL\jhil application2.exe

```
===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 1
Enter Roll No: 1
Enter Name: jhil
Enter Course: bscit
Enter Marks: 85
Student added successfully!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 1
Enter Roll No: 2
Enter Name: vrunda
Enter Course: bscit
Enter Marks: 70
Student added successfully!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 2
Roll    Name    Course  Marks   Grade
```

C:\JHIL\jhil application2.exe

```
===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 2
Roll    Name    Course  Marks   Grade
-------------------------------------------
1       jhil    bscit   85      A
2       vrunda  bscit   70      B

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 3
Enter Roll No to search: 2
2       vrunda  bscit   70      B

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Marks
0. Exit
Enter choice: 4
Enter Roll No: 2
Enter new marks: 83
Marks updated!

===== STUDENT RECORD MANAGEMENT =====
1. Add Student
```

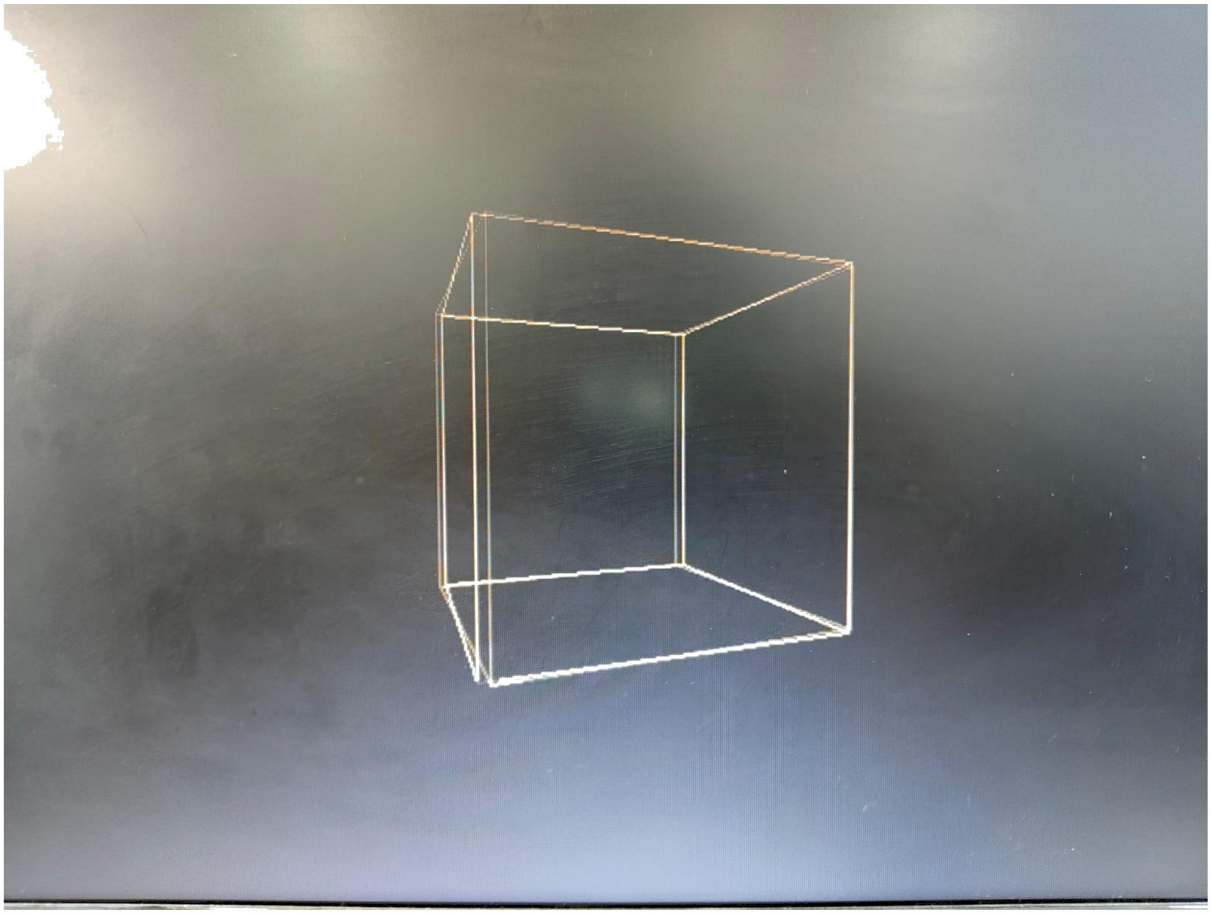
2. **Design a 3D Model using c++ graphics.**

```
[■]                    \TURBOC3\PROJECTS\GRAPHIC.CPP                    1=[↑]
#include <graphics.h>
#include <conio.h>
#include <math.h>                    SS
#include <dos.h>

#define PI 3.14159265

void main()
{
    int gd = DETECT, gm;

    // Change path if your Turbo C++ folder is different
    initgraph(&gd, &gm, "C:\\TURBOC3\\BGI");

    int size = 80;
    int xc = 320;
    int yc = 240;

    float angle = 0;

    while (!kbhit())
        3:30
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```

```
≡  File  Edit  Search  Run  Compile  Debug  Project  Options  Window  Hel
[■]                    \TURBOC3\PROJECTS\GRAPHIC.CPP                    1=[↑]

    while (!kbhit())
    {
        cleardevice();

        float x[8] = {-size, size, size, -size, -size, size, size, -size};
        float y[8] = {-size, -size, size, size, -size, -size, size, size};
        float z[8] = {-size, -size, -size, -size, size, sisze, size, size};

        float x2d[8], y2d[8];

        int i;
        for (i = 0; i < 8; i++)
        {
            float xRot = x[i]*cos(angle) - z[i]*sin(angle);
            float zRot = x[i]*sin(angle) + z[i]*cos(angle);

            float distance = 300;
            float scale = distance / (distance + zRot);

            x2d[i] = xc + xRot * scale;
        27:60
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```

```
File  Edit  Search  Run  Compile  Debug  Project  Options  Window  Help
[■]  \TURBOC3\PROJECTS\GRAPHIC.CPP  1=[↑]
            x2d[i] = xc + xRot * scale;
            y2d[i] = yc + y[i] * scale;
        }

        // Draw Front Square
        line((int)x2d[0], (int)y2d[0], (int)x2d[1], (int)y2d[1]);
        line((int)x2d[1], (int)y2d[1], (int)x2d[2], (int)y2d[2]);
        line((int)x2d[2], (int)y2d[2], (int)x2d[3], (int)y2d[3]);
        line((int)x2d[3], (int)y2d[3], (int)x2d[0], (int)y2d[0]);

        // Draw Back Square
        line((int)x2d[4], (int)y2d[4], (int)x2d[5], (int)y2d[5]);
        line((int)x2d[5], (int)y2d[5], (int)x2d[6], (int)y2d[6]);
        line((int)x2d[6], (int)y2d[6], (int)x2d[7], (int)y2d[7]);
        line((int)x2d[7], (int)y2d[7], (int)x2d[4], (int)y2d[4]);

        // Connect Front and Back
        line((int)x2d[0], (int)y2d[0], (int)x2d[4], (int)y2d[4]);
        line((int)x2d[1], (int)y2d[1], (int)x2d[5], (int)y2d[5]);
        line((int)x2d[2], (int)y2d[2], (int)x2d[6], (int)y2d[6]);
        line((int)x2d[3], (int)y2d[3], (int)x2d[7], (int)y2d[7]); S
60:68
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```

```
File  Edit  Search  Run  Compile  Debug  Project  Options  Window  Help
[■]  \TURBOC3\PROJECTS\GRAPHIC.CPP  1=[↑]

        // Draw Back Square
        line((int)x2d[4], (int)y2d[4], (int)x2d[5], (int)y2d[5]);
        line((int)x2d[5], (int)y2d[5], (int)x2d[6], (int)y2d[6]);
        line((int)x2d[6], (int)y2d[6], (int)x2d[7], (int)y2d[7]);
        line((int)x2d[7], (int)y2d[7], (int)x2d[4], (int)y2d[4]);

        // Connect Front and Back
        line((int)x2d[0], (int)y2d[0], (int)x2d[4], (int)y2d[4]);
        line((int)x2d[1], (int)y2d[1], (int)x2d[5], (int)y2d[5]);
        line((int)x2d[2], (int)y2d[2], (int)x2d[6], (int)y2d[6]);
        line((int)x2d[3], (int)y2d[3], (int)x2d[7], (int)y2d[7]); S

        angle += 0.05;

        delay(30);
    }

    closegraph();
    getch();
}                                                          S
69:69
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```



Explantion:

This program creates and displays a rotating 3D cube using the BGI graphics library in Turbo C++. The cube is defined using eight vertices in 3D space with x, y, and z coordinates. A rotation transformation is applied around the Y-axis using trigonometric functions (sine and cosine) to make the cube spin continuously. Since the computer screen is two-dimensional, perspective projection is used to convert the 3D coordinates into 2D screen coordinates, creating a depth effect where distant points appear smaller. The cube is drawn as a wireframe using line functions, and the animation continues in a loop until a key is pressed, demonstrating basic concepts of 3D graphics, rotation matrices, and real-time animation.